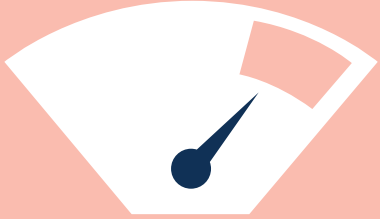




The Scientific Method

At the heart of the process of science is the scientific method. The scientific method follows these steps:

Identify a problem or set of observations.



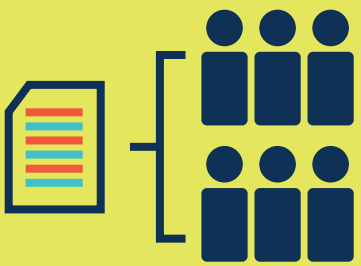
Scientists identify a problem or set of observations that needs an explanation. This represents the purpose for doing a scientific study. For example, scientists may observe that people spontaneously eat less and lose weight on low carbohydrate diets. However, the scientists don't know whether it is the reduction in carbohydrates, or the increase in protein, that is responsible for the decrease in food intake.

Formulate a hypothesis.

Scientists formulate a hypothesis. A hypothesis is a proposed explanation for the problem or set of observations. In the example from step #1, a scientist might hypothesize that the increase in protein intake is responsible for the decrease in calorie intake, due to protein's appetite suppressive effects.



Design a study to test the hypothesis.



A study is designed to test the hypothesis. Scientists select the materials and methods for the study and carry it out. They develop a set of predictions of things that should happen in the study, if the hypothesis is true. In the example, the scientists divide people into two groups. One group lowers carbohydrate intake while keeping protein intake low. The other group lowers carbohydrate intake while increasing protein intake. The scientists track food intake and body weight over time.

Collect, synthesize, and interpret data.

Scientists collect data, synthesize it, and interpret it in light of the hypothesis. They find that the data either supports the hypothesis to varying degrees or does not support it. In the example, if the hypothesis is true, then the group that increased protein intake should end up eating less, and losing more weight, than the group that only decreased carbohydrate intake.



Discard or change the hypothesis if the data does not support it.



If the data doesn't support the hypothesis, it's changed or discarded. In the example, if both groups decreased calorie intake the same, this would suggest the reduction in carbohydrate was responsible for the reduction in appetite, rather than the increase in protein.

OR

Continue testing data that supports the hypothesis.

If the data supports the hypothesis, it will continue to get tested. Scientists will perform more studies using better methods. Other independent scientists may try to replicate the research. The results of further studies will either strengthen or challenge the findings.

